SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

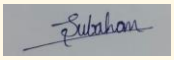
CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Student Name:	T. Subahan	**Reg. No.:**	192511121
Date:	22/10/26	**Duration:**	60 minutes

Code of Conduct

I T. Subahan (192511121) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:
Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**
   - Identify all primary and secondary actors interacting with the system.
   - List the major use cases associated with each actor.
   - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction
   - Problem Description

- Scope of the System
- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
   - Verify whether all stakeholder requirements have been addressed.
   - Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential

(15%)	sections.			components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

Software Requirement Specification (SRS)

Vehicle Service and Maintenance Management System

Scenario-Based Requirement Analysis Software Engineering Assignment

1. Study and Analyse the Scenario

1.1 Scenario Overview

The Vehicle Service and Maintenance Management System is a software application designed to simplify and automate vehicle servicing activities. It allows vehicle owners to book service appointments, maintain vehicle service records, receive maintenance reminders, and track the status of their vehicles during servicing.

The system connects vehicle owners, service center staff, service technicians, and administrators through a centralized platform. It reduces manual paperwork, prevents missed maintenance schedules, improves communication, and helps service centers manage their daily operations efficiently.

1.2 Business Domain

- Domain: Automobile and vehicle maintenance services.
- Application type: Web-based or mobile-based management system.
- Primary purpose: To automate vehicle servicing, appointment scheduling, maintenance tracking, and service history management.

1.3 Stakeholders

Stakeholder	Role and Responsibility
Vehicle Owner	Registers vehicles, books appointments, checks service history, and tracks service status.
Service Center Staff	Manages appointments, confirms bookings, and assigns technicians.
Service Technician	Inspects vehicles, performs maintenance, and updates repair progress.
Administrator	Manages users, service centers, service categories, and system settings.
Notification Service	Sends appointment confirmations and maintenance reminders.
Service Center Manager	Monitors service operations, technician workload, and completed services.

1.4 Existing Challenges

1. Manual appointment booking can result in scheduling conflicts.
2. Paper-based service records may be lost or damaged.
3. Vehicle owners may forget periodic maintenance schedules.
4. Customers may need to call the service center repeatedly to check service progress.
5. Service centers may find it difficult to allocate technicians and manage workloads.
6. Lack of centralized information can lead to delays and communication errors.

1.5 Objectives of the System

- Provide convenient online service appointment booking.
- Maintain accurate and accessible vehicle service histories.
- Send timely maintenance reminders based on dates, mileage, or service intervals.
- Allow vehicle owners to monitor service progress.
- Help service centers manage bookings and technician assignments.
- Improve transparency, efficiency, and customer satisfaction.
- Protect customer and vehicle information through appropriate security controls.

2. Identify Functional and Non-Functional Requirements

2.1 Functional Requirements

Functional requirements describe what the system must do.

ID	Functional Requirement
FR-01	The system shall allow users to register, log in, log out, and manage their profiles.
FR-02	The system shall allow vehicle owners to add, view, update, and remove their vehicle details.
FR-03	The system shall display available service centers, service types, and appointment slots.
FR-04	The system shall allow vehicle owners to book service appointments by selecting a vehicle, service type, date, and time.
FR-	The system shall allow users to reschedule or cancel appointments

05	according to service center policies.
FR-06	The system shall allow service center staff to confirm, reject, and manage appointment requests.
FR-07	The system shall allow staff to assign technicians to service jobs.
FR-08	The system shall allow technicians to update service status, including inspection, in progress, waiting for parts, completed, and ready for collection.
FR-09	The system shall maintain service history, including service date, work performed, replaced parts, and maintenance notes.
FR-10	The system shall send appointment confirmations, status updates, and maintenance reminders.
FR-11	The system shall generate maintenance reminders using configured service intervals, dates, or recorded mileage.
FR-12	The system shall allow vehicle owners to view the current service status of their vehicles.
FR-13	The system shall allow authorized staff to record service charges and provide an itemized service summary.
FR-14	The system shall allow administrators to manage users, service centers, service categories, and system configurations.
FR-15	The system shall generate reports on appointments, completed services, pending jobs, and service center performance.
FR-16	The system shall maintain role-based access so that users can access only the functions permitted for their roles.

2.2 Non-Functional Requirements

Non-functional requirements describe how well the system must operate.

Category	**Requirement**
Performance	Common pages should load within 3 seconds under normal operating conditions.
Security	Passwords must be securely hashed, sensitive data protected in transit, and access controlled by role.
Reliability	The system should prevent duplicate bookings for the same service slot and preserve valid records.

Usability	The interface should be simple, consistent, and suitable for users with basic computer knowledge.
Scalability	The architecture should support increasing numbers of users, vehicles, appointments, and service centers.
Maintainability	The system should use modular components, documented APIs, and automated tests where practical.
Availability	The target availability should be at least 99.5% per month, excluding planned maintenance.
Data Integrity	Vehicle, appointment, and service history records must remain consistent when created or updated.
Compatibility	The application should work on current versions of commonly used web browsers and mobile devices.
Backup and Recovery	The system should perform regular backups and support restoration after data loss.
Privacy	Personal information should be collected and retained only as needed and handled according to applicable privacy laws.
Accessibility	Forms, buttons, labels, and status indicators should be accessible and easy to understand.

3. Identify Actors and Use Cases

3.1 Primary and Secondary Actors

Primary actors directly use the system to achieve their goals.

- Vehicle Owner
- Service Center Staff
- Service Technician
- Administrator
- Service Center Manager

Secondary actor:

- Notification Service — supports the application by sending email, SMS, or in-app notifications.

3.2 Major Use Cases

Actor	Use Cases
Vehicle Owner	Register/Login, Manage Profile, Add Vehicle, Book Appointment, Reschedule/Cancel Booking, View Service History, Track Service Status, Receive Reminders, View Service Charges
Service Center Staff	View Bookings, Confirm/Reject Appointment, Allocate Service Slot, Assign Technician, Record Service Charges
Service Technician	View Assigned Jobs, Inspect Vehicle, Update Service Status, Record Repairs, Update Parts and Maintenance Notes, Complete Service
Administrator	Manage Users, Manage Service Centers, Manage Service Categories, Configure System Settings, Generate Reports
Service Center Manager	Monitor Appointments, Review Technician Workload, View Service Reports
Notification Service	Send Booking Confirmation, Send Status Notifications, Send Maintenance Reminders

3.3 Use Case Diagram

The following diagram represents the interactions between actors and the Vehicle Service and Maintenance Management System.

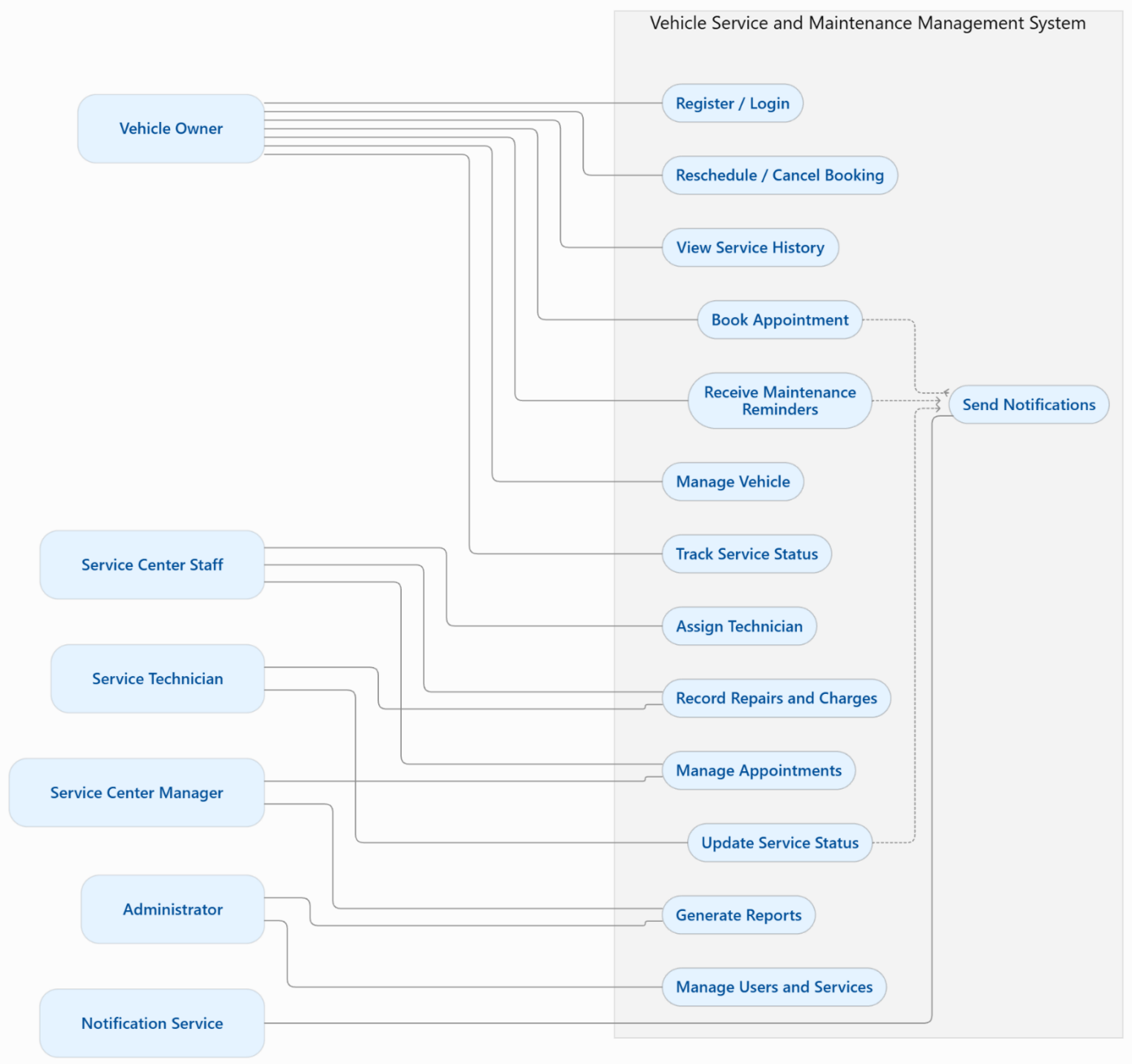


Diagram explanation: The vehicle owner manages bookings and checks maintenance information. Service staff organize appointments and technician assignments. Technicians update the work progress. Administrators manage the platform, while managers monitor service center performance. The notification service delivers reminders and updates.

3.4 Use Case Descriptions

Use Case 1: Book a Service Appointment

Field	Description
Use Case ID	UC-01
Primary Actor	Vehicle Owner
Objective	Book a vehicle service appointment.
Preconditions	The owner is logged in and has registered a vehicle.
Trigger	The owner selects the Book Appointment option.
Main Flow	1. The owner selects a vehicle. 2. The system displays available service types. 3. The owner selects the required service. 4. The system displays available dates and time slots. 5. The owner selects a slot. 6. The system validates availability. 7. The system saves the booking and generates a booking ID. 8. The system sends a confirmation.
Alternative Flow	If the slot is unavailable, the system asks the owner to select another slot. If required details are missing, the system requests them.
Postconditions	The appointment is saved and its status is set to Pending or Confirmed, according to the booking policy.

Use Case 2: Track Service Status

Field	Description
Use Case ID	UC-02
Primary Actor	Vehicle Owner
Objective	Check the progress of a vehicle service.
Preconditions	The owner is authenticated and has access to the relevant booking.
Main Flow	1. The owner opens My Appointments. 2. The system displays the owner's bookings. 3. The owner selects an appointment. 4. The system displays the current status, service details, and latest update.
Alternative Flow	If no booking exists, the system displays an appropriate message.

Postconditions	The owner receives the latest available service information.

Use Case 3: Update Service Status

Field	Description
Use Case ID	UC-03
Primary Actor	Service Technician
Objective	Record the progress of service work.
Preconditions	The technician is logged in and assigned to the service job.
Main Flow	1. The technician opens assigned jobs. 2. The technician selects a vehicle. 3. The technician records inspection findings. 4. The technician updates the status and maintenance notes. 5. The system saves the update. 6. The system notifies the owner when appropriate.
Alternative Flow	If the technician is not authorized for the job, the system denies the update.
Postconditions	The service record contains the latest status and progress details.

Use Case 4: Receive Maintenance Reminder

Field	Description
Use Case ID	UC-04
Primary Actor	Vehicle Owner
Supporting Actor	Notification Service
Objective	Remind the owner about upcoming vehicle maintenance.
Preconditions	Vehicle information and a reminder schedule are available.
Main Flow	1. The system checks the maintenance schedule. 2. It identifies upcoming or overdue maintenance. 3. It prepares a reminder. 4. The notification service sends the reminder through the configured channel.
Alternative Flow	If notification delivery fails, the system records the failure and retries according to its notification policy.
Postconditions	The reminder attempt is logged, and its delivery status is recorded when available.

4. Software Requirement Specification (SRS) Document

4.1 Introduction

The Software Requirement Specification defines the functional and non-functional requirements, actors, system interfaces, data requirements, and constraints of the Vehicle Service and Maintenance Management System.

This document serves as a common reference for developers, testers, project managers, service center staff, and other stakeholders during software design, implementation, and validation.

4.2 Problem Description

Many vehicle service centers rely on phone calls, paper records, or separate systems to manage appointments and service histories. These methods can lead to scheduling conflicts, missing records, delayed maintenance, and poor visibility into repair progress.

Vehicle owners may not know when their next service is due or whether their vehicle has completed servicing. Service center employees may also struggle to coordinate bookings, technicians, and parts.

The proposed system addresses these issues by providing a centralized platform for appointment management, maintenance tracking, service history, and notifications.

4.3 Scope of the System

In scope:

- User registration, login, and profile management.
- Vehicle registration and information management.
- Service center and service type selection.
- Appointment booking, rescheduling, and cancellation.
- Technician assignment and service progress tracking.
- Maintenance history and reminders.
- Service charge recording and summaries.
- Role-based access and administrative reports.
- Email, SMS, or in-app notifications through configured services.

Out of scope for the initial version:

- Advanced vehicle diagnostics through onboard vehicle sensors.
- Direct integration with every vehicle manufacturer's systems.

- Insurance claim processing.
- Spare-parts procurement and supplier management.
- Full accounting, payroll, and inventory management.
- AI-based failure prediction.

These features may be considered in future versions if stakeholders require them.

4.4 System Features

Feature	Description	Priority
User Management	Registration, login, profiles, and permissions	High
Vehicle Management	Add and maintain vehicle information	High
Appointment Management	Book, confirm, reschedule, and cancel appointments	High
Service Tracking	View and update repair progress	High
Service History	Store and retrieve completed service records	High
Maintenance Reminders	Send date- or mileage-based alerts	High
Notifications	Inform users about booking and status changes	Medium
Service Charges	Record charges and display a service summary	Medium
Reporting Dashboard	View booking and service performance reports	Medium
Advanced Analytics	Analyze service patterns and predict maintenance needs	Low / Future

Recommended development order: Implement user management, vehicle management, booking, service tracking, and history first. Add reminders and reporting after the core workflow is reliable.

4.5 Data Requirements

The system must store and manage the following information.

Entity	Important Attributes
User	User_ID, Name, Email, Phone, Password_Hash, Role
Vehicle	Vehicle_ID, Owner_ID, Registration_Number, Make, Model, Year, Mileage
Service_Center	Center_ID, Name, Address, Contact_Number, Operating_Hours
Service_Type	Service_ID, Service_Name, Description, Estimated_Duration, Base_Cost
Appointment	Appointment_ID, Vehicle_ID, Owner_ID, Center_ID, Service_ID, Date, Time, Status
Technician	Technician_ID, Name, Contact_Number, Specialization
Service_Record	Record_ID, Appointment_ID, Technician_ID, Inspection_Notes, Work_Performed, Parts_Used, Cost, Completion_Date
Maintenance_Schedule	Schedule_ID, Vehicle_ID, Service_Type, Due_Date, Due_Mileage, Reminder_Status
Notification	Notification_ID, User_ID, Message, Channel, Sent_Time, Delivery_Status
Payment/Charge	Charge_ID, Appointment_ID, Amount, Description, Payment_Status

Relationships between entities:

- One user can own multiple vehicles.
- One vehicle can have multiple appointments.
- One service center can handle multiple appointments.
- One appointment can have one or more service records or progress updates.
- One technician can be assigned to multiple service jobs over time.
- One vehicle can have multiple maintenance schedules.
- One user can receive multiple notifications.

Data validation rules:

1. Every vehicle must be linked to a valid owner.
2. Each appointment must reference a registered vehicle and valid service center.

3. The system must prevent conflicting confirmed bookings for the same service slot and capacity.
4. Service status changes must follow an approved workflow.
5. Service charges must be non-negative, and the currency must be clearly specified.
6. Only authorized users may view or modify restricted customer and service information.
7. Important updates should be timestamped and associated with the user or staff member who performed them.

4.6 External Interface Requirements

A. User Interface

- Login and registration pages.
- Vehicle management dashboard.
- Service booking form and appointment calendar.
- Appointment and service history pages.
- Technician work dashboard.
- Administrator dashboard and reports.
- Notification and reminder settings.

B. Hardware Interface

- A computer, smartphone, or tablet with a supported browser.
- A server or cloud-hosted environment for running the application.
- An internet connection for online access.

No direct vehicle hardware integration is required in the initial version.

C. Software Interface

- Database management system such as MySQL or PostgreSQL.
- Web application server and API layer.
- Email or SMS gateway for notifications.
- Optional payment gateway if online payment is introduced.
- Authentication service for secure login, if externally managed.

D. Communication Interface

- HTTPS for secure client-server communication.
- REST APIs or equivalent interfaces for communication between application components.
- Secure API authentication and authorization.

- Appropriate error responses and logging for failed requests.

4.7 Assumptions

1. Vehicle owners have access to a smartphone or computer and an internet connection.
2. Users provide valid contact information and accurate vehicle details.
3. Service centers maintain updated operating hours and appointment availability.
4. Technicians update service progress accurately and promptly.
5. The system has access to a functioning database and hosting environment.
6. Users have appropriate permissions based on their assigned roles.
7. A third-party notification provider is available if email or SMS notifications are enabled.
8. Maintenance reminders use configured schedules and recorded mileage; the system does not automatically read the vehicle's odometer in the initial version.

4.8 Constraints

Constraint Type	Description
Technical	The application depends on network connectivity, database availability, and compatible devices.
Operational	Staff must be trained to update appointments and service records consistently.
Economic	Hosting, database, maintenance, SMS, and other third-party service costs must remain within the project budget.
Legal	Personal data must be handled according to applicable privacy, security, and consumer-protection requirements.
Time-related	Development and testing must be completed within the academic project deadline.
Integration	External services may impose API limits, subscription fees, or availability restrictions.
Data-related	Accurate maintenance reminders depend on the quality of the recorded vehicle and service data.

5. Validate Requirement Completeness and Consistency

Requirement validation checks whether the SRS is clear, complete, consistent, feasible, and testable.

5.1 Completeness Checklist

SRS validation checklist

0 of 9

User registration and role-based access

FR-01 and FR-16

Vehicle registration and management

FR-02

Booking, rescheduling, cancellation, and slot validation

FR-03 to FR-06

Technician assignment and service progress

FR-07 and FR-08

Service history and maintenance reminders

FR-09 to FR-12

Service charges and management reports

FR-13 to FR-15

Security, performance, reliability, and scalability

Non-functional requirements

Data entities, relationships, and interfaces

Data and external interface requirements

Assumptions, constraints, and validation

Sections 4.7, 4.8, and 5

5.2 Ambiguity, Redundancy, and Inconsistency Checks

Potential Issue	Validation or Improvement
Unclear meaning of "service completed"	Define valid status values and the transitions between them.
Unspecified booking confirmation	Specify whether bookings are automatically confirmed or require staff approval.
Duplicate appointments	Validate slot capacity when confirming a booking, including simultaneous requests.
Unclear reminder timing	Define reminder rules, such as a configurable number of days before the due date.
Unclear service charges	Specify who enters charges, whether estimates can change, and how final costs are recorded.
Duplicate customer and vehicle records	Apply suitable uniqueness checks, such as registration number within the relevant jurisdiction.
Inconsistent permissions	Define permissions for each actor and enforce them in the application and API.
Unspecified notification failure	Log failed deliveries and define retry and fallback behavior.

5.3 Suggested Requirement Improvements

To make the SRS more precise and suitable for testing, the project team should:

1. Assign a unique identifier to every requirement.
2. Use the word "shall" for mandatory system behavior.
3. Define measurable performance and availability targets.
4. Specify booking cancellation deadlines and rescheduling policies.
5. Define service status transitions, for example: Pending → Confirmed → In Progress → Completed → Ready for Collection.
6. Define retention periods, backup frequency, and recovery targets.
7. Map each requirement to one or more test cases.
8. Review the requirements with vehicle owners, service staff, technicians, and administrators before implementation.

5.4 Sample Acceptance Criteria

Requirement	Testable Acceptance Criteria
Appointment Booking	Given an available slot, when a registered owner submits valid booking details, then the system creates one appointment and displays its booking ID.
Slot Conflict Prevention	Given a slot that has reached its capacity, when another owner attempts to book it, then the system rejects the conflicting booking and offers other available slots.
Service Tracking	Given an authorized owner with an active appointment, when they open the appointment details, then the system displays the latest saved status.
Maintenance Reminder	Given a vehicle with a configured upcoming maintenance date, when the reminder condition is met, then the system creates a reminder and records the notification attempt.
Access Control	Given a user without technician permissions, when they attempt to update a service job, then the system denies the operation.

6. Overall Requirement Validation and Conclusion

6.1 Requirement Quality Assessment

Quality Attribute	Validation Result
Completeness	Core functions for owners, staff, technicians, and administrators are included.
Consistency	Requirements are aligned with the system objectives and data model.
Clarity	Requirements use defined actions, actors, and expected outcomes.
Feasibility	The core system can be developed using standard web technologies and a relational database.
Testability	Functional requirements have corresponding measurable acceptance criteria.

Traceability	Requirement IDs can be linked to use cases, implementation modules, and test cases.

The requirements are suitable as a baseline SRS for an academic project. Before implementation, the team should confirm operational policies, notification rules, security requirements, and performance targets with the relevant stakeholders.

6.2 Conclusion

The Vehicle Service and Maintenance Management System provides a centralized solution for managing vehicle servicing activities. The scenario-based analysis identifies the major stakeholders, functional and non-functional requirements, actors, use cases, data entities, interfaces, assumptions.

The proposed SRS establishes a foundation for system architecture design, database design, implementation, and software testing. By automating appointment booking, maintaining service histories, providing maintenance reminders, and enabling service status tracking, the system can reduce manual effort, improve operational efficiency, and enhance customer satisfaction.